HACKNATION 2026 · CHALLENGE 3
Databricks Agentic Healthcare Maps

AarogyaNet

Guide care, don't just map it.

10,000

facilities ingested, cleaned, and trust-scored
across India — every recommendation grounded in real Databricks data.

THE PROMISE

Google Maps shows the nearest hospital. Nearest is not best.
AarogyaNet triages the patient, ranks hospitals on calibrated trust,
books bed + ambulance + doctor + drug atomically, and learns from outcomes.

3,736	**4-tier**	**$0.30**	**0.831 → 0.350**
PINs mapped	Trust badge	LLM cost / 256	Trust learns

LIVE STACK

Backend · https://aarogyanet-api-production.up.railway.app
Frontend · https://app-wine-pi.vercel.app
Workspace · Databricks dbc-17e9d40d-5056
Sponsor · Vector Search · FM APIs · Genie · MLflow Agent Bricks

EXECUTIVE SUMMARY

One paragraph, six numbers, one promise.

AarogyaNet is a multi-agent healthcare map that triages a patient in plain language (Hindi, Urdu, English voice or text), ranks nearby hospitals on a calibrated trust score derived from a two-model LLM consensus, and books bed, ambulance, doctor, and drug as a single atomic transaction. After care, patient outcomes feed back into the trust score in real time. Density does not equal access — and a single LLM is not a guardrail. AarogyaNet treats both as engineering problems, not slogans.

10,000 Facilities ingested and cleaned	**3,736** Indian PINs mapped on NGO Desert Map	**4** Atomic resources per booking saga	**$0.30** Total LLM cost for 256 facilities
0.831 → 0.350 Trust drift after 6 outcomes	**4-tier** Trust badge (rule / single / verified / disagree)	**20** Foundation Model endpoints live	**3** Apps on one data layer

WHAT MAKES THIS A PRODUCT, NOT A DEMO

A closed feedback loop: triage → ranked recommendation → atomic booking → outcome ledger → recalibrated trust. Most hackathon demos stop after the recommendation step. AarogyaNet writes the loop end-to-end.

01 · THE PROBLEM

Maps are not enough. Lives are.

Every minute of delay in a cardiac event raises mortality by ~1%. Indian patients face four compounding failures the moment they search for care:

WRONG DESTINATION

Google Maps returns the nearest hospital — not the best one. Nearest can be full, lack the right specialist, or be unsafe. Distance is the only signal.

HIDDEN TRUST GAP

Public listings mark a clinic as Level-1 trauma; the equipment log has been silent since 2023. Single-source claims hide stale, unverifiable, or contradictory facts.

BOOKING IS THEATRE

A 'booked' bed without a confirmed ambulance, doctor, or drug is a promise the system cannot keep. Patients arrive to half-fulfilled reservations every day.

NO LEARNING LOOP

The system never asks: did the patient survive, recover, get transferred? Trust is published once and never updated. Yesterday's rating decides tomorrow's referral.

These are not UX problems. They are data, distributed-systems, and feedback-loop problems. AarogyaNet treats them as such.

02 · TARGET AUDIENCE

Three users. Three apps. One data layer.

The same Databricks lakehouse powers three distinct surfaces, each tuned to a different decision-maker. Building three products from one pipeline is the platform-maturity story for the judges.

PRIMARY · PATIENT

Aradhna · 34 · Mumbai

Chest pain at midnight. Speaks Hindi. Needs the right cardiology bed, an ambulance, a doctor, and the right drug — guaranteed before she leaves home.

SECONDARY · CLINICIAN

Dr. Iyer · ER lead

Triaging a referral inbound from a smaller clinic. Needs trust badges, ETA, capacity at receiving hospitals, and an audit trail for the handoff.

TERTIARY · POLICY / NGO

State Health NGO Analyst

Wants to ask, in plain English, ‘which PINs in Bihar have zero oncology coverage?’ — and get a SQL-grounded answer to fund the right new clinic.

HOW WE SERVE EACH

Surface	Who	Job to be done	Killer affordance
PatientFlow	Patient	Get the right care now	Voice triage + atomic 4-resource booking
DoctorCopilot	Clinician	Refer with confidence	Trust badge + SSE reasoning stream
NGODashboard	NGO / policy	See the deserts	Per-PIN heatmap + Genie NL→SQL queries

03 · SOLUTION CORE FEATURES

Seven pillars. One closed loop.

Each feature ships as a standalone endpoint and a UI surface. Together they form the only loop that matters: triage, rank, book, learn.

01

Two-model trust consensus

Llama 3.3 70B (Extractor) and Llama 4 Maverick (Validator) score every facility on four sub-factors — bed availability, oxygen, drug stock, specialist on-call. Agreement-band logic produces a 4-tier badge: **rule / single-source / verified / disagreement**. Disagreements are surfaced, never silently downgraded. Total LLM cost for 256 facilities ≈ $0.30.

02

Atomic 4-resource booking saga

txn_atomic reserves bed + ambulance + doctor + drug across four resource tables in a single transaction. If any leg fails, compensating actions roll back all four. The demo includes a deliberate 'ambulance unavailable' failure — every tile rolls back live in the UI. No half-promised care.

03

Outcome learning loop

/outcome records a patient ping after care. A **v_trust_calibrated** view recomputes trust on the fly. The demo arc shows Aradhna's clinic at 0.831 → 0.350 after six negative outcomes. The AI does not just recommend — it learns.

04

Vernacular voice triage (Hindi / Urdu)

Web Speech API on the patient side captures Hindi or Urdu speech. After /book commits, Fish Audio TTS narrates the dispatch in the same language. Inclusivity is wired into the protocol, not bolted on.

05

Genie — natural-language SQL for NGOs

POST /sponsor/genie/query wraps Databricks Genie. An NGO analyst types 'Bihar PINs with zero oncology' and receives executable SQL plus rows from **gold_pin_capabilities**. Live mode calls **WorkspaceClient().genie**; canned mode returns a deterministic, brief-cited fallback so the demo never 5xx's. Six pre-curated queries cover Bihar appendectomy access, oncology deserts, two-model disagreements, ICU availability, calibrated trust ≥ 0.80, and cardiology rankings.

06

NGO Desert Map

Per-PIN capability heatmap over 3,736 Indian PINs across four specialties (Trauma, Cardiac, Neuro, Pediatric). The headline: **Bihar — 149 PINs with zero oncology, 130 with zero emergency. Maharashtra — 1,492 facilities, 403 PINs with zero oncology.** Density is not access.

07

SSE reasoning stream

Every agent decision streams over **/sse** as a structured event: *triage → extractor → validator → router*. The frontend renders the chain in real time. **/sse_demo** serves a pre-recorded fallback if the live channel falters mid-pitch.

04 · UNIQUE SELLING PROPOSITION

What only AarogyaNet does.

Most hackathon healthcare demos stop at ‘LLM gives advice.’ AarogyaNet ships the discipline of a production system in three places where competitors cut corners.

Dimension	Typical hackathon entry	AarogyaNet
LLM trust	Single LLM call, taken at face value	Two-model consensus, 4-tier confidence badge, disagreement surfaced not hidden
Booking	‘Reserve bed’ button writes one row	Atomic saga across 4 resource tables with compensating-action rollback
Feedback	No outcome capture	Patient outcome ledger that recalibrates trust on the fly
Inclusivity	English-only chat	Hindi / Urdu voice input + TTS narration in same language
NGO surface	Often missing	Second product on the same data — Desert Map + Genie NL→SQL
Demo robustness	Live API or bust	**SAFE_DEMO=true** kill-switch, /sse_demo fallback, Genie canned mode, smoke test on every deploy

Trust is not a single LLM call. It is a two-model consensus, with a four-tier confidence badge, recalibrated from real patient outcomes — and the booking that follows is atomic, or it is rolled back. That is the AarogyaNet wedge.

05 · IMPLEMENTATION & TECHNOLOGY

Architecture, stack, deployment.

DATA AND CONTROL FLOW

```
Patient (Hindi / Urdu voice or text)
      |
      v
  /triage    --> Vector Search BGE-large + Llama 3.3 fallback
      |          (specialty, urgency, fast_path, red_flag_match)
      v
  /recommend --> Router: P(bed) x travel x specialty x trust
      |          Trust = Llama 3.3 + Llama 4 Maverick consensus
      |          Badge: rule / single / verified / disagreement
      |          SSE: triage -> extractor -> validator -> router
      v
  /book      --> Atomic saga: bed + ambulance + doctor + drug
      |          txn_atomic; ANY leg fails  ->  rollback all 4
      |          hash_patient_id before warehouse insert
      v
  /outcome   --> Patient ping after care
      |          v_trust_calibrated view recomputes trust
      v
  Demo arc   :   Aradhna 0.831 -> 0.350 after 6 negatives

  /sponsor/genie/query  --> Databricks Genie  ·  NL -> SQL
                            live: WorkspaceClient().genie
                            canned: deterministic demo fallback
```

STACK

Layer	Technology	Why
Frontend	React 18 · Vite · TypeScript · Tailwind · Leaflet	Map-first UX with live SSE reasoning panel
Backend	FastAPI · slowapi · SSE · Pydantic	Async + rate-limited; SSE without extra infra
Data	Databricks SQL Warehouse · Vector Search (BGE-large)	Symptom corpus + facility text indexes
LLMs	Llama 3.3 70B · Llama 4 Maverick · GPT-5.x fallback	Two-model trust consensus, ~$0.30 / 256 facilities
Agent	MLflow ResponsesAgent (Mosaic AI Agent Bricks)	Sponsor conformance, traceable agent runs
NL→SQL	Databricks Genie · WorkspaceClient.genie SDK	Plain-English queries on gold tables for NGOs
Voice	Web Speech API (in) · Fish Audio TTS (out)	Hindi / Urdu inclusivity, post-booking narration
Logs	Token-scrubbing LogRecordFactory · MLflow tracing	No **dapi*** / **sk-*** token ever reaches a handler
Hosting	Railway (backend) + Vercel (frontend)	Autodeploy on main; smoke script on every deploy

DEMO-SAFETY ENGINEERING

Every sponsor route is flag-gated and default-off. **SAFE_DEMO=true** is a master kill-switch — every sponsor endpoint then returns a pre-baked artifact from **app/sponsor/_demo/**. Genie has its own canned mode with deterministic, brief-cited responses. **/sse_demo** replays a recorded reasoning stream when live SSE falters. **scripts/smoke_railway.sh** validates **/health**, **/triage**, **/sse**, **/book** after every Railway deploy.

SECURITY POSTURE

X-Demo-Key auth on /book and /outcome. **PII_SALT** hashes patient_id before any warehouse insert (deterministic but non-reversible). A global **LogRecordFactory** redacts **dapi***, **sk-***, and Fish-key shapes from every log line and traceback before they hit any handler. Failing-test-first discipline: every known bug has a matching test in **tests/test_known_bugs_*.py** that flips RED → GREEN as the fix lands.

06 · RESULTS & IMPACT

What is on the table, what it changes.

Concrete artefacts, concrete metrics. Everything below is live and verifiable today.

LIVE METRICS

20 Foundation Model endpoints exposed	**4** Specialties on Desert Map	**256** Facilities two-model verified	**6** Genie queries brief-cited
3,736 PINs mapped with severity tags	**4-tier** Trust badge resolution	**100%** /health uptime on Railway	**$0.30** LLM consensus cost / 256 facilities

DEMO ARC THAT SELLS THE LOOP

PATIENT JOURNEY · 60 SECONDS

Hindi voice → triage classifies cardiology, urgency 4, fast_path → map shows 3 ranked hospitals with **verified** and **disagreement** badges → reserve clicked → 4 tiles light up → ambulance fails → all four tiles roll back live. No half-promise.

OUTCOME LOOP · THE LEARNING MOMENT

Switch to Doctor Copilot. Aradhna's clinic, trust 0.831 last week. Six negative patient outcomes recorded. Trust now 0.350. The next patient will not be routed there. The system learns in real time.

NGO MOMENT · GENIE NL → SQL

Type ‘Bihar PINs with zero oncology’ into the Genie panel. Output: executable SQL, 5 rows from **gold_pin_capabilities**, headline **97.4% of Bihar PINs uncovered**, ~104M people affected. Density is not access.

07 · TECHNICAL VIDEO PRESENTATION

Full narration script — 90 seconds.

Read straight to camera. Times are cumulative. Every claim has a live endpoint or warehouse table behind it.

Time	Beat	Narration
0:00 — 0:08	**HOOK · hold on the patient screen**	“When someone has a heart attack in Mumbai, Google Maps shows the nearest hospital. Nearest is not best. AarogyaNet guides patients to the right care at the right time.”
0:08 — 0:22	**PATIENT DEMO · voice triage**	Speak Hindi into the patient screen. Voice transcribes, /triage returns specialty=cardiology, urgency=4, fast_path=true, red_flag_match=[chest pain]. The map renders three ranked hospitals with trust badges — verified, single-source, disagreement. Cost and ETA visible on each card.
0:22 — 0:38	**ATOMIC SAGA · the drama**	Click reserve. Four tiles light up in sequence: bed ✓, doctor ✓, drug ✓, ambulance ✗. All four tiles roll back live. “If even one resource fails — we do not promise the patient anything we cannot deliver.”
0:38 — 0:52	**OUTCOME LOOP · trust learns**	Swipe to Doctor Copilot. “This clinic was trust 0.831 last week. Six negative patient outcomes later — 0.350. Trust learns in real time. Most demos cut off here. Ours does not.”
0:52 — 1:08	**NGO + GENIE · the second product**	Switch to NGO Dashboard. “Maharashtra has 1,492 facilities — and 403 PINs with zero oncology. Density is not access.” Open the Genie panel. Type ‘Bihar PINs with zero oncology.’ Genie returns the SQL, the rows, the headline: 97.4% uncovered, 104M people.
1:08 — 1:22	**TRUST CONSENSUS · the moat**	Hold on a disagreement badge. “Llama 3.3 said 0.80 on bed availability. Llama 4 Maverick said 0.00. We do not silently average. We surface the disagreement and route it for human review. Two-model consensus, four-tier badge, ~$0.30 of LLM cost for 256 facilities.”
1:22 — 1:30	**CLOSE · one line**	“Three apps, one Databricks lakehouse, one closed loop: triage, rank, book, learn. **AarogyaNet — guide care, don't just map it.**”

DELIVERY NOTES FOR THE TECHNICAL VIDEO

Speak slow on the rollback (0:30 — 0:38). The four-tile sequence is the visual centerpiece — let it breathe. On the Genie segment, type the question into the live UI rather than narrating over a still: judges remember the moment SQL appears under a Hindi place-name. Keep the closing line on the screen for two beats before fade.

08 · CLOSING

One tagline. One promise. One repo.

“Guide care, don't just map it.”

AarogyaNet is the loop the brief asks for: triage, rank, book, learn. Three apps, one Databricks lakehouse, every recommendation grounded in calibrated trust, every booking atomic, every outcome fed back into the next decision.

LIVE API	https://aarogyanet-api-production.up.railway.app
WEB APP	https://app-wine-pi.vercel.app
WORKSPACE	Databricks · dbc-17e9d40d-5056 · warehouse 10fff96dd6d936b5
STACK	Databricks Vector Search · Foundation Model APIs · Genie · SQL Warehouse · MLflow Agent Bricks
TEAM	AarogyaNet · 4 contributors · HackNation 2026

Built for HackNation 2026 · Challenge 3 · Databricks Agentic Healthcare Maps.